

PAIRS TRADING IN KOREA · 2026

Pairs Trading in Korea

Korean Stock Pairs and KOSPI200 Cash–Futures

Tax · short-selling limits · structural pairs. Three inputs change the conclusions.

Stocks

Cash–futures

Rules

Verdict

Examples are hypothetical, built from simulation and published rule values

Overview

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EXECUTIVE SUMMARY

Same formulas, three different inputs, different conclusions

Modern Pairs Trading (ten techniques), recomputed under Korean market conditions.

Sell-side tax 0.20% from Jan 2026 — round-trip cost more than twice the US level

Short-selling limits Only the 350 KOSPI200 + KOSDAQ150 names; candidate pairs shrink to 1.9%

Structural pairs 105 common-preferred pairs come without any cointegration search

Conclusion Priority: ⑦ optimal band → ⑩ DSR → ⑥ regime → ② sparsity

Techniques that price in costs and rules come before techniques that refine the signal

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Tax · short selling · structural pairs

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01. What is different

Three inputs change the strategy design

The two on the left hurt; the one on the right helps.

Sell-side tax**0.20%**

Transaction tax 0.05% + rural tax 0.15%. The US is effectively zero

Raised Jan 2026**Short-sale universe****350 names**

13.6% of 2,570 names. Only 1.9% of pairs work both ways

Index members only**Structural pairs****105 pairs**

Common and preferred share one cash-flow claim

No search needed

Tax widens the band, short-sale limits shrink the universe, structural pairs remove multiple testing

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01. What is different

Every backtest built on 2025 tax rates is wrong

Applies to trades executed from 2 January 2026.

Market	Through 2025	From 2026	Round-trip change
KOSPI market	Rural tax 0.15%	Transaction tax 0.05% + rural tax 0.15% = 0.20%	+10bp
KOSDAQ market	Transaction tax 0.15%	Transaction tax 0.20%	+10bp

- **Annual impact** A 20-day half-life pair turned 12 times a year misses 1.2%p a year
- **Threshold impact** The cash-futures arb threshold rises 17%, from 3.20p to 3.76p
- **What to do** Make cost parameters date-dependent and rerun the backtest

Arbitrage lives near the cost threshold, so it is the most tax-sensitive

02

Stocks

How to trade common against preferred shares

Structural pairs · mean estimation error · cost-aware band

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02. Stocks · A-1

Same cash flows, so no cointegration test

They differ only in voting rights and dividend priority.

$$g_t = \frac{P_t^c - P_t^p}{P_t^c}, \quad dg_t = \kappa(\bar{g} - g_t) dt + \sigma dW_t$$

- **Candidates** Keep preferreds trading $\geq$ KRW 1bn a day
- **Price gap** 3+ years of closes; mark ex-dividend and rights-issue dates
- **Stationarity** Test with ADF. No rejection, drop the pair
- **Hedge ratio** No regression — 1:1 value-neutral
- **Execution order** Fill the thinner preferred leg first

Beta has little room to vary, so a Kalman filter adds little

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02. Stocks · A-1

Three years of data put the half-life short, at 21 days

750 days simulated from an OU with true mean 15.0%, sd 2.5%p, half-life 25 days.

Item	Estimate	True	Band	Signals / yr
Mean gap	14.27%	15.00%	$ z > 1.0$	12.7
Stationary sd	2.46%p	2.50%p	$ z > 1.5$	5.0
Half-life	21.2 d	25.0 d	$ z > 2.0$	2.3
ADF p-value	0.0081	—	—	—

A short half-life means a short holding period — and an understated stock-borrow fee

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02. Stocks · A-2

The effective sample is reversion cycles, not days

The band is measured from the mean — and the mean has sampling error.

$$\widehat{g} = \frac{\hat{a}}{1-\hat{b}}, \quad \text{se}(\widehat{g}) \approx \sigma_{\text{eq}} \sqrt{\frac{1}{\kappa T}}$$

- **Step 1** Compute the effective sample kappa·T per pair. Distrust it below 30
- **Step 2** Express the mean's standard error in sigma units and compare with the band
- **Step 3** If the ratio exceeds 20%, push the band out by that much
- **Step 4** If more data does not help, drop the pair from the universe

A narrow band around an unknown centre is random trading

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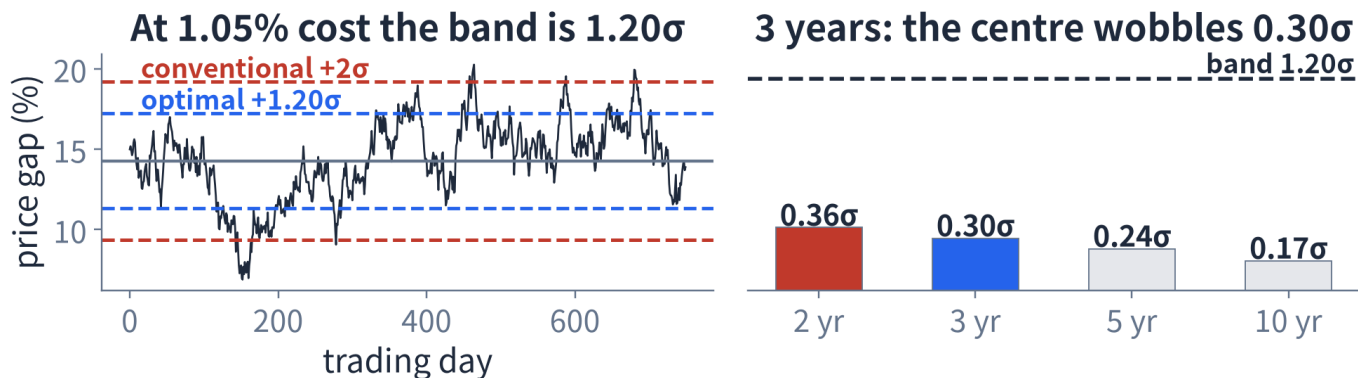
Cash-futures

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02. Stocks · A-2

With three years of data the band centre wobbles 0.30σ

 0.36σ

2 yr

Effective n 14

 0.30σ

3 yr

Effective n 21

 0.24σ

5 yr

Effective n 35

 0.17σ

10 yr

Effective n 69

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02. Stocks · A-3

Only the stock-borrow fee scales with holding time

Stack it leg by leg to see which side to cut.

$$c_{rt} = \underbrace{2f + \tau_s + 2i_p}_{\text{long leg}} + \underbrace{2f + \tau_s + 2i_c}_{\text{short leg}} + \ell \frac{h}{250}$$

$$\ell_{\text{eff}} = \ell \cdot \frac{h}{250}, \quad h = \frac{\ln 2}{\kappa}$$

- **Impact** Preferreds cost 3–4× commons. Regress on fill data
- **Borrow cost** Even at 2.5% a year: 21bp at a 21-day half-life, 60bp at 60 days
- **Units** Express cost in sigma, not %, to compare with the band

If the gain is under 1.2×, band optimisation is not worth the effort

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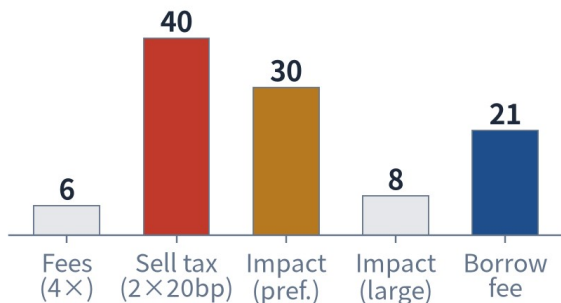
Rules

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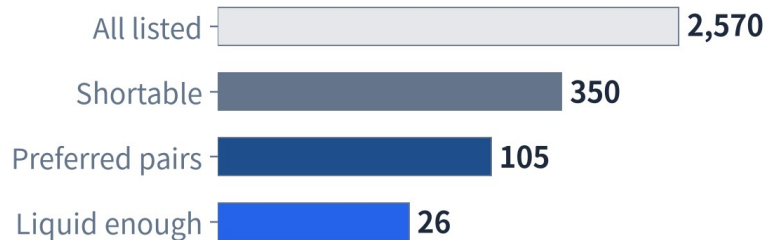
02. Stocks · A-3

105bp round trip — two-thirds is tax and market impact

105bp round trip, 2/3 tax & impact



26 tradable structural pairs

**105bp**Round-trip cost
= 0.43σ**1.20σ**Optimal band
Gap 17.22%**1.49×**vs conventional
±2σ baseline**0.90σ**No short selling
Directional exposure left

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02. Stocks · A-3

Without short selling the cost falls, but the strategy changes

The 52.2bp short leg disappears — and the full market exposure stays.

$$b^* = \arg \max_b \frac{b - c_{rt} / \sigma_{eq}}{\mathbb{E}[T_{-b \rightarrow 0}] + \mathbb{E}[T_{0 \rightarrow b}]}$$

Both legs available

- 105bp round trip, optimal band 1.20σ
- Market-neutral — index direction irrelevant

No short selling

- 53bp round trip, optimal band 0.90σ
- Gap narrows, market falls — you still lose

Long-only gap trading is value investing, not pairs trading

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02. Stocks · A-4

The rules cut the candidates by 98%

The rules do the job of ③ clustering in Modern Pairs Trading (ten techniques) before it starts.

Stage	Names	Possible pairs	Share
All listed (KOSPI 820 + KOSDAQ 1,750)	2,570	3,301,165	100%
Shortable (KOSPI200 + KOSDAQ150)	350	61,075	1.9%
Common-preferred structural pairs	—	105	—
Preferred daily value $\geq$ KRW 1bn	—	26	—

All 26 pairs share the same factors, so the number of independent bets is far smaller

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02. Stocks · A-4

Short selling was fully banned for 3 of the last 6 years

Before any backtest: what will you do when it is banned?

Period	Measure	Effect on pairs
2020.3 ~ 2021.5	Full ban (COVID response)	No short leg
2021.5 ~ 2023.11	Resumed, index members only	Universe of 350
2023.11 ~ 2025.3	Full ban (about 17 months)	No short leg
2025.3.31 ~	Resumed (index members only)	Universe of 350

Alternatives: sell single-stock futures, sell from existing holdings, or pause the strategy

03

Cash-futures

KOSPI200 arbitrage is not risk-free

Fair value · cost threshold · dividend error · rollover

Overview

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03. Cash-futures · B-1

The signal is the gap, not the basis

When dividends exceed interest, a negative basis is normal.

$$F_{t,T}^* = S_t e^{r(T-t)} - D_{t,T}, \quad D_{t,T} = \sum_{t < t_i \leq T} d_i e^{r(T-t_i)}$$

$$\text{Basis}_t = F_t - S_t, \quad \text{Gap}_t = F_t - F_{t,T}^*$$

- **Rate** 91-day CD if borrowing; short treasuries if opportunity cost
- **Dividends** Weight expected dividends by index share; reinvest to expiry
- **Replication** If a top-100 proxy tracks worse than the gap, it is a bet

The market basis is observed; the gap is a judgement

Overview

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Cash-futures

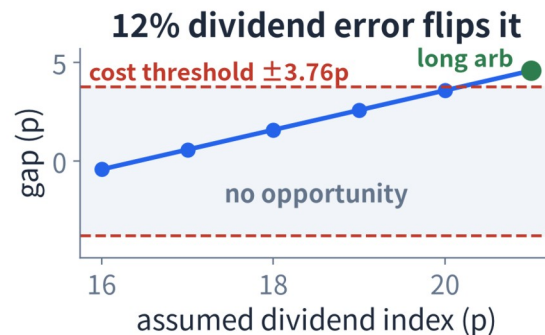
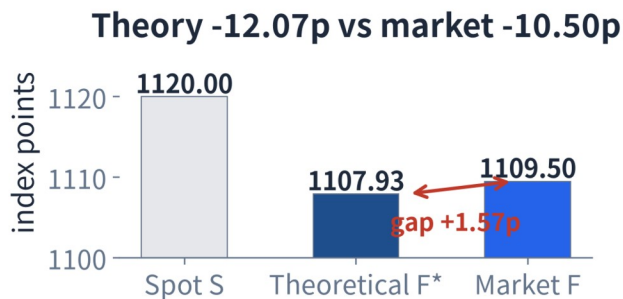
Rules

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03. Cash-futures · B-1

The +1.57p gap vanishes once costs are included

Assumes spot 1,120.00, 91-day CD 3.21%, 60 days to expiry, expected dividend index 18.00p.

**1107.93****Fair value F*** $S \cdot e^{-rt} - D$ **1109.50****Market F**

Observed

+1.57p**Gap**

+0.141%

3.76p**Cost threshold**

Not tradable

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03. Cash-futures · B-2

Long and short arbitrage thresholds are not symmetric

Short arbitrage must borrow the basket, which adds a borrow cost.

Long arb: $\text{Gap}_t > +c_{rt} \implies$ buy basket, sell futures

Stage	Component	bp
Buy spot basket	Fee 1.5 + impact 5.0	6.5
Sell spot basket	Fee 1.5 + tax 20.0 + impact 5.0	26.5
Sell and buy back futures	0.3×2	0.6
Round-trip total	3.76p at index 1,120	33.6

Long arbitrage pays no tax on entry and pays it once on exit

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03. Cash-futures · B-3

A 100bp rate error equals a 1.84p dividend error

Rates are observed; dividends must be estimated.

$$\frac{\partial \text{Gap}}{\partial D} = +1, \quad \frac{\partial \text{Gap}}{\partial r} = -S(T - t)e^{r(T-t)}$$

- **Ranges** Build optimistic and pessimistic cases, not a point
- **Whole range** Clearing the threshold only at the centre is not enough
- **Year-end** Dec and Mar contracts carry large dividend error; Jun and Sep small
- **Calendar** Fair value jumps around earnings and dividend announcements
- **Post-check** Log realised vs estimated dividend index and recalibrate

Dividends feed fair value and are random — so this is not riskless arbitrage

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03. Cash-futures · B-3

A 12% dividend error reverses the conclusion

Market price fixed at 1,109.50; only the dividend index assumption varies.

Dividend index	Fair value F*	Gap	Verdict
16.00	1,109.93	-0.43p	No opportunity
18.00 (base)	1,107.93	+1.57p	No opportunity
20.00	1,105.93	+3.57p	No opportunity
20.19 (break-even)	1,105.74	+3.76p	Threshold reached
21.00	1,104.93	+4.57p	Long arb

Held to expiry, realised dividends settle it; closed early, the error becomes P&L

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03. Cash-futures · B-4

Roll cost is the deviation from the theoretical spread

The theoretical spread is carry — time value, not cost.

$$C_{\text{roll}} = n_{\text{roll}} \cdot \mathbb{E} \left[F^{\text{next}} - F^{\text{near}} - \left(F^{*,\text{next}} - F^{*,\text{near}} \right) \right]$$

0.40p

Slippage per roll
On spread quotes

4

Rolls per year
Mar · Jun · Sep · Dec

1.60p

Annual roll cost
0.143% of index

0.43%

Hurdle return
Below it, a third is eaten

Filling the legs separately carries the index move in between — trade on spread quotes

04

Rules

Sometimes the stop-loss cannot work

Price limits · sidecar · circuit breaker · quadruple witching

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04. Rules · B-5

Stop-loss rules must not assume they can be filled

Rules	Trigger	Effect on pairs
Price limit	$\pm 30\%$ of prior close	A stock locked limit-up cannot be covered
Sidcar	KOSPI200 futures $\pm 5\%$ for 1 minute	Program-trading orders halted 5 minutes
Circuit breaker, level 1	Index -8% for 1 minute (declines only)	20-minute trading halt
Quadruple witching	2nd Thursday of Mar, Jun, Sep, Dec	Program volume piles into the closing auction

Modern Pairs Trading (ten techniques): ㉞ and ㉟ assume every order fills at your price

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04. Rules · B-5

Write the can't-fill branch into the rules

A price-only stop-loss does nothing at limit-up.

- | | | |
|---|-----------------------------------|--|
| 1 | Substitute hedge | If the short leg is blocked, use single-stock or index futures |
| 2 | No entry under sidecar | While program orders are halted, only one leg may fill |
| 3 | Ignore signals near expiry | Gaps within 3 business days of witching are not mean-reversion signals |
| 4 | Forced exit | If a failed fill breaches a limit, exit at the next day's open |

If the stop cannot fill at the band, the computed return is never realised

05

Verdict

Which of the ten techniques survive in Korea

Verdicts and priorities · execution checklist

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05. Verdict

Verdicts in Korea: selection and dynamics techniques

Technique	Verdict	Reason
① PCA factor residuals	Limited	350 shortable names make factor estimates unstable
② Sparse mean-reversion	Useful	Fewer legs cut tax and impact directly
③ Clustering filter	Unneeded	The rules already cut the universe 98%
④ Kalman varying beta	Useful	Little help for preferreds; useful for sector pairs
⑤ Copula signals	Not advised	No edge in Modern Pairs Trading (ten techniques); adds cost
⑥ Regime detection	Essential	Short-sale bans and sidecars are regime switches

In Korea the rules take over the multiple-testing control of ③ clustering

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Verdicts in Korea: timing and validation techniques

Technique	Verdict	Reason
⑦ Cost-aware band	Most important	Costs are double, so band optimisation pays more
⑧ RL band	Not advised	Its fill assumption clashes with the rules
⑨ Deep-learning signal	Not advised	Lost to z-score in Modern Pairs Trading (ten techniques)
⑩ Deflated Sharpe	Essential	Picking from 26 candidates, the trial count is the overfit

Do first

- ⑦ optimal band → ⑩ DSR
- ⑥ regime detection → ② sparsity

Skip

- ⑤ ⑧ ⑨ signal-refinement family
- Bands that ignore costs and rules

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06. Checklist

Six checks at the design stage

Miss one and the backtest will not match live trading.

- **Tax rate** Is the 2026 sell tax of 0.20% applied (not 0.15%)?
- **Borrow** Is the stock-borrow fee scaled by holding period?
- **Impact** Is impact for preferreds and small caps measured separately?
- **Sample** Is the effective sample $\kappa \cdot T$ at least 30?
- **Est. error** Is the mean's standard error compared with the band width?
- **Availability** Are the shortable list and borrow availability checked?

All six are about costs and constraints — none is about the signal

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06. Checklist

Checks for arbitrage and operations

Cash-futures arbitrage

- Dividend index estimated as a range?
- Threshold cleared across the whole range?
- Separate long and short arb thresholds?
- Replication error smaller than the gap?

Operations

- Stop-loss has a can't-fill branch?
- No-entry rule for sidecars and expiry?
- Plan for a short-selling ban?
- Trials counted and corrected with DSR?

Reference values

Transaction tax effective 2026.1.2 · 91-day CD 3.21% (2026.9.22) · KOSPI200 futures 1,127.75 (2026.9.23)

Rules change. For every backtest, check the tax rate and short-sale access in force

PAIRS TRADING IN KOREA

In Korea, costs and rules come before the signal

Apply in order: ⑦ optimal band · ⑩ DSR · ⑥ regime detection · ② sparsity